

TUNNELING PAIRS

The Josephson effect is a rare example of a quantum phenomenon observable on a macroscopic scale. It involves an electric current flowing indefinitely, without an applied voltage, across a Josephson junction. A Josephson junction is a pair of superconductors with a thin barrier sandwiched between them. Electrons in Cooper pairs (see p.81) tunnel through the barrier from one superconductor to the other with no resistance. Josephson junctions are used in SQUIDS (see pp.96–97).

Across the barrier

Pairs of electrons tunnel through the barrier from one superconductor to the other until a critical current is reached and a voltage appears across the junction.

BARRIER

SUPERCONDUCTOR

THIN BARRIER

A very thin barrier of insulating or normal conduction material between two superconductors forces the pair to tunnel (see pp.70–71) across.

